SNA – Concluding Lecture

From Networks to Power Systems

Dr Dhiren Patel

15 April 2026

What have we really learned?

- **Move beyond definitions**
- Graph ≠ data structure
- Graph = **representation of reality**
- SNA ≠ analysis
- SNA = **control layer over systems**

- **If you can map the network, you can predict it. If you can predict it, you can influence it.**

Interpretation (math on the next slide)

Component	**Meaning**
Structure (G)	Who is connected
Signals (edges, entropy)	What information flows
Intervention (T)	What you change

Equations to Rule All Concepts

- CNA: $Y_i = f\left(T_i, T_{N(i)}, G\right)$ (page 15 from last deck)

- ERGM: $P(G) \propto \exp\left(\theta^T s(G)\right)$ (page 55)

- Exposure mapping: $E_i = \frac{\sum_{j \in N(i)} T_j}{|N(i)|}$ (page 19)
- Unify them conceptually:
- **Reality = f (Structure + Signals + Interventions)**

Constructing the SNA Stack

- Everything we studied falls into 4 layers:
- First, structure — who is connected
- Then, signal — which edges actually matter
- Then, identity — who is really who
- And finally, dynamics — how to change the system
- // Most real-world failures happen not in structure…
  but in identity and signal layers
- // Failures usually originate in Layer 2 (signal) and Layer 3 (identity)

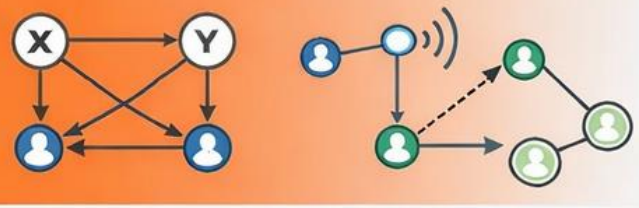
The SNA Stack

Power of Networks

From Mapping
→ To Control

Dynamics & Intervention

- CNA (Causal Effects)
- Diffusion: Shockwave



Identity

- FONDUE Framework
- Entity Resolution

Signal Quality

- Deep Link Entropy
- Weak vs Strong Edges

- Deep Link Entropy
- Weak vs Strong Edges

Structure

- Graph topology
- Communities, Centrality

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More interpretation - The 4-Layer SNA Stack

Layer 1: Structure	Layer 2: Signal Quality	Layer 3: Identity	Layer 4: Dynamics & Intervention
• Graph topology • Communities, centrality	• Deep Link Entropy • Weak vs strong edges	• FONDUE • Entity resolution	• CNA (causal effects) • Diffusion (shock-wave) • Network interventions

Core insight: Most systems fail not at structure—but at identity and signal layers

Same Mathematics, Different Domains

Let’s see how this same structure explains every domain

Domain	**Node**	**Edge**	**Intervention**
Epidemic	Person	Contact	Vaccination
Social media	User	Follow/share	Content injection
Finance	Account	Transaction	Blocking / AML (Anti Money Laundering)

Core insight: Domains change; Network logic does not

Wikipedia (Truth as a Network)

- Truth = **network consensus + governance**
- **Connected to:**
- AI alignment
- Decentralized systems
- Blockchain identity
- Wikipedia = human consensus network with editorial governance edges
- AI alignment = constraining output within a graph of acceptable knowledge boundaries
- "AI alignment = graph governance problem"

Truth as a Network (cont.)

- Example - Sam Altman's first mover fiasco with releasing biased ChatGPT without sufficient safety testing (Nov. 2022 – Nov. 2023)
- Microsoft - emphasising products over safety
- Some of the most controversial features, including the ability to fine-tune the model for specific uses and to deploy it as a personal assistant, had not actually received safety approval.
- Sam Altman / ChatGPT alignment controversy - demonstrating that **truth and identity validation are network governance problems**, not just technical ones.
- high-profile example of **SNA Stack failure at Layers 2 and 3 -** *The "fiasco" = what happens when signal quality (Layer 2) and identity/scope (Layer 3) lack sufficient guardrails*
- AI Alignment (containing output within desired scope with required guardrails).

Power of Network - Where SNA dominates the world

- Future systems will not be databases. They will be **graphs with policies**

- Cybersecurity → attacker graphs
- Blockchain → identity graphs
- AI → knowledge graphs
- Governance → population graphs

Where SNA Fails

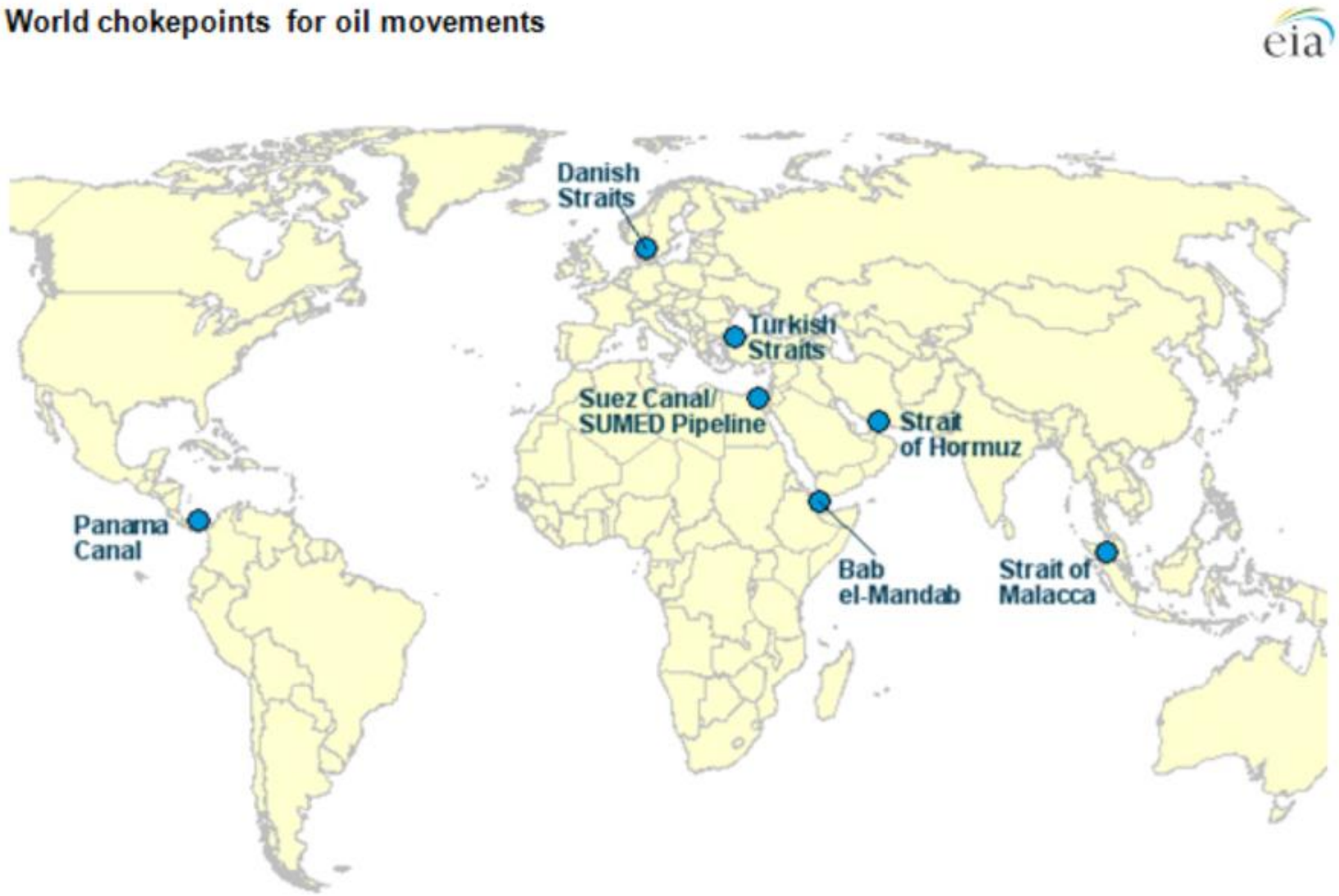
- The biggest threat is not wrong analysis; It is missing edges.

- Data incompleteness
- Adversarial manipulation
- Non-stationarity
- Hidden edges

Case Study – Straight of Hormuz

- A. Why non-soverign currency Bitcoin is getting upward price pressure due to this geopolitical issue
- B. Why USA is failing in spite of being global military super power!! (SNA concepts applied)

Strait of Hormuz is chokepoint for 20% of world's oil



Source: U.S. Energy Information Administration.

A. Hormuz toll tax - the *institutionalization* dimension (10 April 2026)

- A war disruption is temporary — markets price in a recovery.
- A toll tax, if it holds, is a structural change to the cost of global trade.
- Every barrel that passes through Hormuz becomes permanently more expensive.

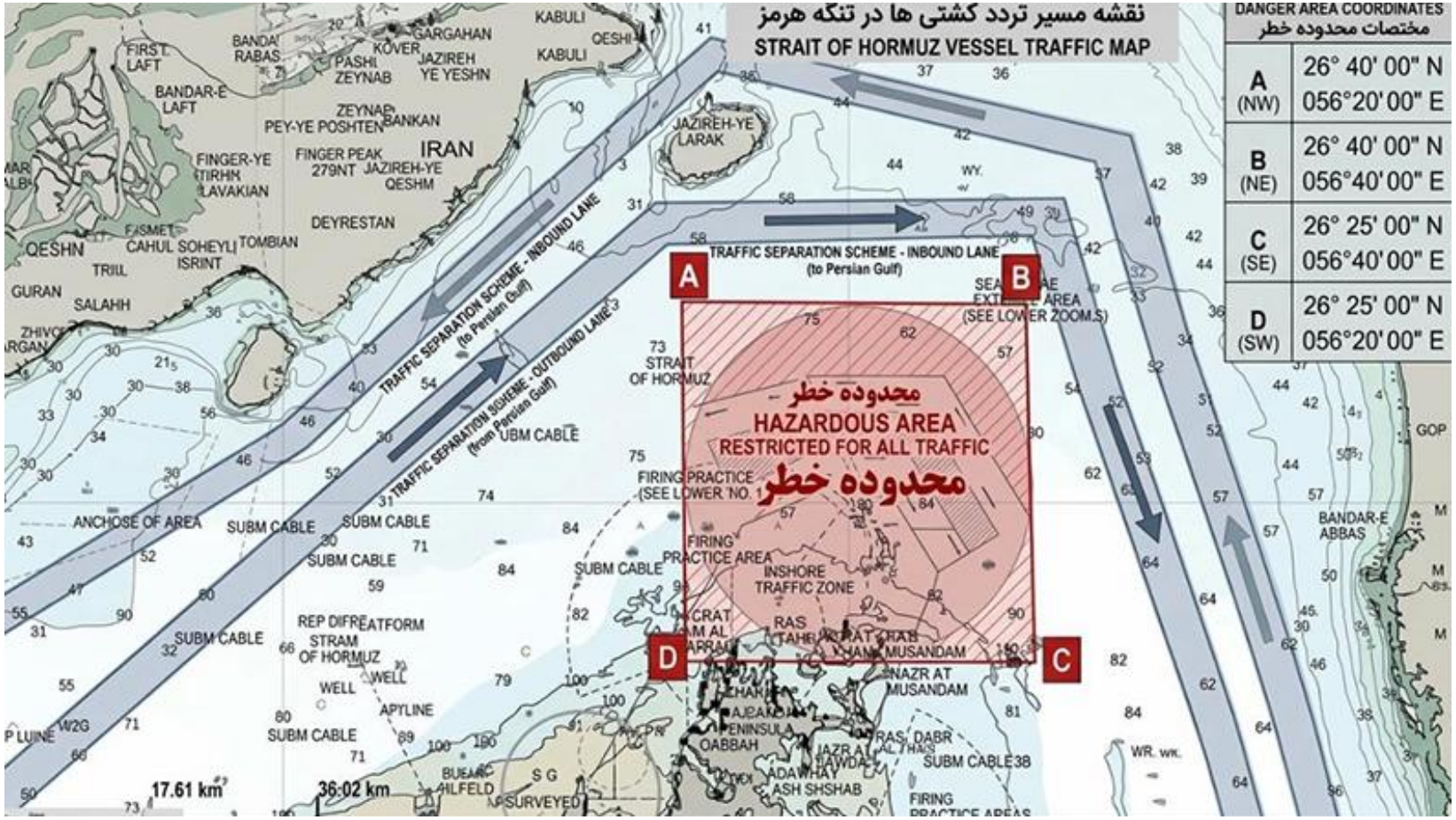
1 USD per Barrel tax (just one dollar!!)

- Normal transits in both directions are around 135 ships a day.
- An estimated 2,000 ships — including oil tankers, bulk carriers, and six cruise liners — remain stranded in the Persian Gulf with around 20,000 seafarers aboard, unable to pass through.
- If we assume a conservative average of around 1 million barrels per oil tanker, and roughly 300–400 oil tankers among the stranded fleet, the crude oil alone locked in the Persian Gulf likely amounts to **300–500 million barrels**
- **400 M USD toll tax…**

Crude oil price chart (10 April) //usd per barrel

- 1 **Barrel** (Oil): ~159 litres
- 1 Barrel (Oil): 42 US gallons
- 1 Barrel (Oil): ~35 Imperial (UK) gallons
- Metric Tonne Relation: 1 cubic meter of oil equals approximately 6.29 barrels.
- Refining Increase: When refined, one 159-litre barrel of crude oil can yield about **170 litres of refined products** (like gasoline and diesel) due to **processing gain**.

Sea mines in Strait of Hormuz



Why? *Sanctions evasion demand*

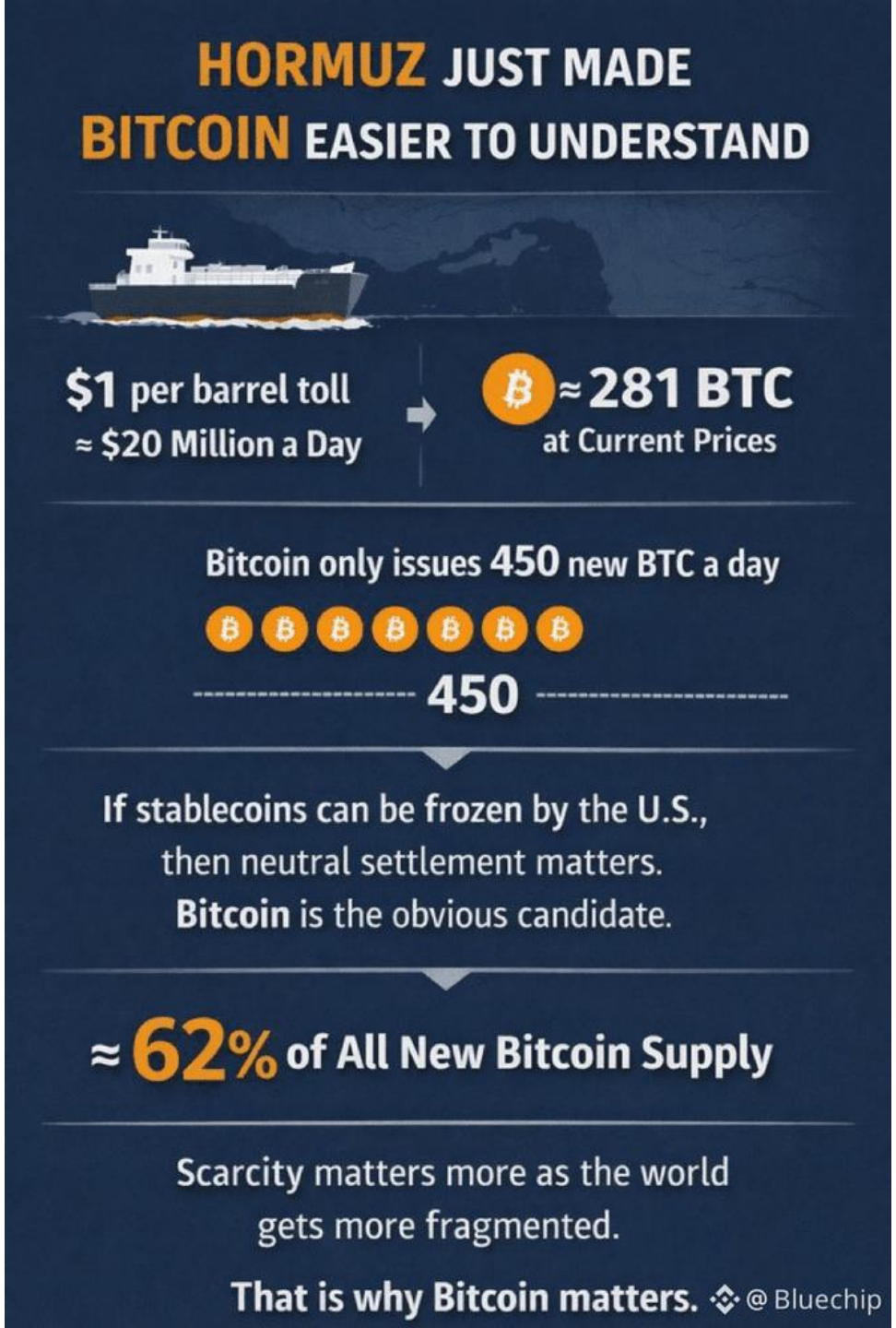
- What? - the US using financial sanctions as a war tool
- Iran, Russia, and sanctioned entities have historically used crypto to settle trade outside SWIFT.
- A formalized toll system creates a parallel need for payment rails outside the dollar system.
- BTC and stablecoins on permissionless chains become more strategically relevant.
- BTC, as a neutral, non-sovereign asset, becomes a credible part of that conversation.
- The thesis is conditional on the crisis being prolonged or the toll tax being formalized into international maritime law

Price pressure on BTC (up)

- 1 block every 10 minutes
- 6 blocks per hour
- 24x6 blocks in a day

= 144 blocks x 3.125 BTC

= 450 BTC



BTC rise <10 April 2026 chart>

BTC price on 15 April

Market Summary > Bitcoin

74,040.41 USD

+1,066.21 (1.46%) ↑ past 5 days

15 Apr, 6:20 am UTC · Disclaimer

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max

B. Problem Framing (Network Perspective)

- We model the geopolitical system as a **multi-layer strategic network**:

- **Nodes (V)**
- **USA** (high military centrality)
- **Iran** (geographically embedded chokepoint controller)
- **EU bloc**: France, UK, Spain, Portugal (political + economic influence)
- **Oil-importing countries** (Asia, others)
- **Shipping routes / commercial vessels** (flow carriers)

Edges (E)

- **Military projection edges** (naval reach)
- **Geographical control edges** (chokepoint proximity)
- **Economic dependency edges** (oil trade)
- **Political alignment edges** (alliances / opposition)

Strait of Hormuz – Marine Traffic

All ship movements in the Strait of Hormus (local time zone). Source: Marine Traffic, Map: ESRI

Apply SNA Concepts

(A) Centrality vs Control (Critical Insight)

- **USA**:
  - High **degree centrality** (global military reach)
  - High **eigenvector centrality** (influence via alliances)
- **Iran**:
  - High **betweenness centrality** (controls chokepoint path)
- **Conclusion:**
- Control of flow depends more on *betweenness* than *degree*

Apply SNA Concepts

- **(B) Edge-Embedded Power (DLE Insight)**
- Using **Deep Link Entropy (DLE)**:
- USA edges:
  - “global military presence” → **high frequency → high entropy → weaker control signal locally**
- Iran edges:
  - “geographic chokepoint + mines + drones” → **rare, localized, low entropy → strong control signal**
- Low-entropy edges (Iran) dominate high-entropy edges (USA)

Apply SNA Concepts

- **(C) Structural Constraint (Graph Topology)**
- Strait of Hormuz behaves like a:
- **Cut-edge / bottleneck**
- Removal or disruption → **global flow collapse**
- Iran sits **on the edge**, not just connected to it.

Apply SNA Concepts

- **(D) Network Intervention Limits**
- USA strategy:
- Intercept ships
- Enforce alternative routes
- Apply economic pressure
- **Why it fails (SNA reasoning):**
- **Edge targeting is incomplete**
  - Cannot remove all Iran-controlled edges
- **High cost of persistent control**
  - Continuous monitoring of dense edge traffic
- **Adversarial adaptation**
  - Mines, drones → **asymmetric edge disruption**

Apply SNA Concepts

- **(E) Multi-Node Resistance (Coalition Dynamics)**
- EU nodes:
- Moderate centrality individually
- But collectively form a **counter-cluster**
- This creates:
- **Negative alignment edges with USA**
- Weakens USA’s ability to enforce global compliance

Apply SNA Concepts

- **(F) Diffusion / Shock-Wave Model**
- If USA enforces control:
- Shock initiates at **policy node (USA)**
- Propagates through:
  - Trade networks
  - Political alliances
- **But:**
- EU resistance = **wave dampening**
- Iran disruption = **wave fragmentation**
- Result: No global propagation → policy fails to stabilize

SNA Stack Interpretation

Layer	Observation
Structure	Iran sits on chokepoint (critical edge position)
Signal	Iran’s actions = low entropy (high impact)
Identity	State actors clear → no ambiguity
Dynamics	Interventions fail due to topology + resistance

Conclusion - B

- In networks with chokepoints, edge-embedded nodes with high betweenness centrality can override nodes with higher global degree centrality.
- USA = **high power node (global influence)**
- Iran = **critical edge controller (local dominance)**
- Therefore:
- "Power in networks is not about how many connections you have, but where you sit in the flow."
- You cannot control a network if you do not control its bottlenecks.

Closing - Final Mental Model

1. Everything is a network
2. Edges matter more than nodes
3. Intervention beats observation

Structure → Signal → Identity → Dynamics

- Computer Science started with computation.
- It is moving toward **connection and control**.
- And networks are the language of that shift…..
- Future systems are not databases; They are **graphs with policies**